Math League Exam 2 Middle Grades
2024-2025 Governor's Cup Competition

1. Which of the following numbers is the largest 5-digit number divisible by 9, 5, 18, 24, and 27?
   a. 98280
   b. 99360
   c. 99900
   d. 99990

2. Simplify the following expression: $12/(\sqrt{98} - \sqrt{108} + \sqrt{50} - \sqrt{48})$. After simplifying and rationalizing this expression, what is the denominator?
   a. 6
   b. 2
   c. 1
   d. 3

3. Express 1.142857142857… in the form of p/q, where p and q are integers and q ≠ 0. Which of the following equals p + q?
   a. 15
   b. 23
   c. 22
   d. 17

4. Consider the nested radical pictured here. Which of the following values best represents X?

$$X = \sqrt{6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \ldots\ldots}}}}$$

   a. 6
   b. 2.8
   c. 3
   d. 4.5

5. The fifth term in an arithmetic sequence is 120. The twelfth term is 162. What is the common difference for this sequence?
   a. 7
   b. 8
   c. 5
   d. 6

6. The first term in a geometric sequence is 1048576. The common ratio is 0.5. What is the $20^{th}$ term in this sequence?
   a. 2
   b. 4
   c. 1/2
   d. 1

7. Your coach prepared 18 1/3 liters of lemonade for an upcoming academic meet. Before the meet started, team members drank 8 1/2 liters of lemonade. After the game, the team drank 1 2/5 less than they drank before the meet. How many liters of lemonade were left after the academic meet?
   a. 2.6
   b. 7/3
   c. 41/15
   d. 8/3

8. Match each of the following properties of equality with the correct example.

| | |
|---|---|
| x. Reflexive | 1. If A = B, then B = A |
| y. Symmetric | 2. A = A |
| z. Transitive | 3. If A = B and B = C, then A = C |

   a. x-1, y-2, z-3
   b. x-2, y-1, z-3
   c. x-3, y-1, z-2
   d. x-1, y-3, z-2

9. Solve the following absolute value problem for x: $|4x - 6| = 8 - 2x$.
   a. 7/3
   b. 7/3 or -1
   c. -7/3 or 1
   d. -1

10. The price of a car was marked at $34,000. At the end of month sale, the dealership lowered the price to $31,500. What is the percent decrease to the nearest percent?
   a. 4%
   b. 96%
   c. 7%
   d. 93%

11. The school band has 98 members. The ratio of girls to boys in the band is 3:4. Which of the following numbers indicates the number of boys in the band?
   a. 42
   b. 56
   c. 14
   d. 52

12. A model of the Empire State Building is built on a scale of 1 cm: 10 feet. If the height of the actual building is 1486 feet, then how tall in meters is the model?
   a. 1.486
   b. 0.1486
   c. 148.6
   d. 14.86

13. Four times the sum of a number and -5 is 4 more than twice the number. What is the number?
a. 5
b. 4
c. 12
d. -5

14. A store's cost for a shirt is $12.50, which the store marked up 364% to sell to customers. During a special sale, the store later sold the shirt with 30% discount. What profit did the store make on this shirt?
a. $29.25
b. $30.60
c. $6.50
d. $19.35

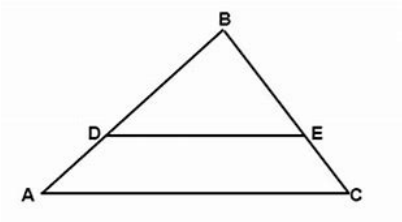
15. Match the following conversions with the correct approximate value.

| | |
|---|---|
| X. 2103 centimeters | 1. 23 yards |
| Y. 18288 millimeters | 2. 18 yards |
| Z. 16.5 meters | 3. 20 yards |

a. X-2, Y-3, Z-1
b. X-3, Y-2, Z-1
c. X-1, Y-3, Z-2
d. X-1, Y-2, Z-3

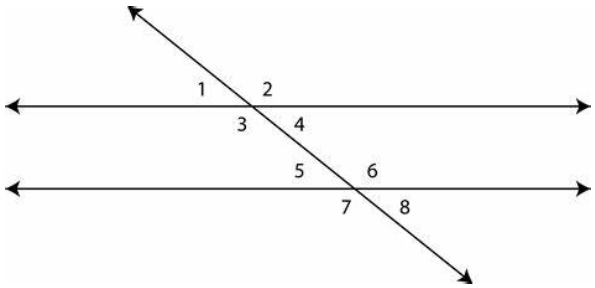
16. A contractor measures the perimeter of a foundation to be 475 feet. When he orders steel, he orders it in hundredths of a meter. How many meters of steel should he order?
a. 144.78 meters
b. 1558.44 meters
c. 434.76 meters
d. 285.92 meters

17. If a person works an eight-hour shift at Governor's Roofing, that person earns $350 dollars. About how much money does he earn each minute he works?
a. $43.75
b. 97 cents
c. $1.23
d. 73 cents



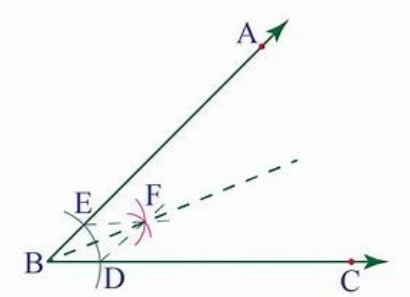

18. Segment DE is parallel to side AC of Triangle ABC. Angle A measures 35° and Angle B measures 78°. What is the measurement in degrees of angle BED?
a. 35°
b. 78°
c. 67°
d. 72.5°

19. Two parallel lines are cut by a transversal. If the angle measure of angle 3 is (6X + 10) ° and the measure of angle 5 is $\left(\frac{36}{11}\right)x°$. Then what is the measure of angle 8?




a. 61.2°
b. 122.2°
c. 60°
d. 120°

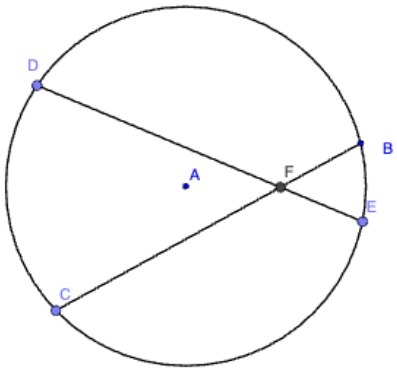
20. Line segment BF is an angle bisector for Angle ABC. The measure of angle EBF is $(4x - 2)°$ and the measure of angle FBD is (x + 12) °. What is the measure of angle ABC?




a. 16.67°
b. 26°
c. 52°
d. 33.33°

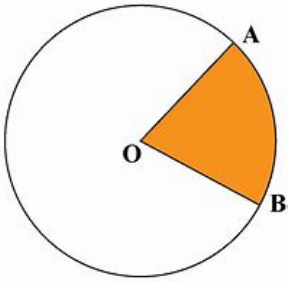
21. How many diagonals does a convex undecagon have?
a. 44
b. 27
c. 54
d. 35

22. The measure of minor arc BE is 28°, and the measure of minor arc DC is 110°. Which of the following degrees indicates the measure of angle BFE?



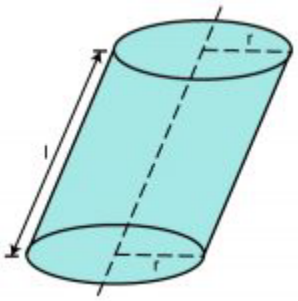

a. 28°
b. 69°
c. 110°
d. 41°

23. The perimeter of a circle is $4\pi$ units. The measure of central angle AOB is 60°. What is the perimeter of the shaded region of the circle?




a. $\frac{2}{3}\pi$
b. $4 + \frac{2}{3}\pi$
c. $\frac{1}{3}\pi$
d. $4\frac{1}{3}\pi$

24. A right triangle is inscribed within a circle having a radius of s units. What is the length of the hypotenuse in units?
a. $\sqrt{2}s$
b. $4s$
c. $2s$
d. $\sqrt{3}s$

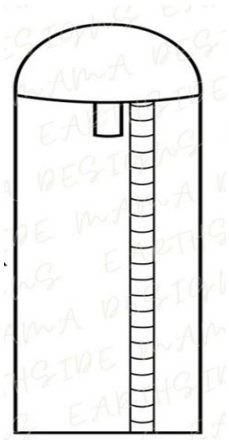


25. What is the total surface area in square units for a cylinder having a slant height of 12 units and a radius of 6 units?
a. $12 + 72\,\pi$
b. $6\sqrt{3} + 72\,\pi$
c. $196\,\pi$
d. $216\,\pi$

26. A sphere has a volume of $2027.8333\pi$ cubic centimeters. What is the radius of this sphere? Answers have been rounded to the nearest tenth.
a. 11.5 centimeters
b. 26.0centimeters
c. 2.3 centimeters
d. 14.5 centimeters

27. A cube has edges measuring 9 inches. If you reduce each edge by 1/3 of its original length, then what is the volume of the reduced cube in cubic inches?
a. 729
b. 81
c. 216
d. 513

28. A silo that consists of a cylinder shape and a hemisphere is 2/3 full of grain. If the cylinder part of the silo has a height of 25 feet and diameter of 10 feet and the hemisphere has a diameter of 10 feet, then how much grain is in the silo? Answers have been rounded to the nearest whole number.
a. 2043 cubic feet
b. 1484 cubic feet
c. 628 cubic feet
d. 2006 cubic feet



29. A right triangle has sides measuring 12 cm and 14 centimeters. What is the hypotenuse for that right triangle rounded to tenths?
a. 15.5 cm
b. 20.7 cm
c. 17.8 cm
d. 18.4 cm

30. A 30°-60° right triangle has a hypotenuse measuring $20\sqrt{6}$ cm. What is the length of the longer leg in centimeters?
a. $10\sqrt{6}$
b. $30\sqrt{3}$
c. $10\sqrt{3}$
d. $30\sqrt{2}$

31. Match the following trigonometric functions with the correct description.

| | |
|---|---|
| X. cosine | 1. Ratio of the adjacent side to the hypotenuse |
| Y. sine | 2. Ratio of the side opposite to the side adjacent |
| Z. tangent | 3. Ratio of the side opposite to the hypotenuse. |

a. X-3, Y-1, Z-2
b. X-1, Y-3, Z-2
c. X-3, Y-2, Z-1
d. X-2, Y-1, Z-3

32. What is the absolute difference between the mean and range for the following data set: 6.5, 4.3, 5, 9.8, 2.2, 7.4?
a. 26/15
b. 5/3
c. 9/5
d. 31/5

33. A data set contains the following values: 4, 7, 5, 5, 6, 7, 8. If you multiply each value by 0.5, what is the new mean?
a. 2.5
b. 4
c. 3
d. 3.5

34. What is the standard deviation for the following data set: 8.8, 6.6, 5.4, 10.3, 9.8, 6.2, and 5.3? Answers have been rounded to the nearest tenth.
a. 2.4
b. 1.9
c. 2.3
d. 1.7

35. In a neighborhood pickup game, each player has 10 shots. Bran has been on a streak making 2/3 of the three-pointers he shoots. If that trend continues, which of the following indicates the probability of making exactly eight 3-pointers in the next game if he shoots ten times.
a. 0.1951
b. 0.0867
c. 0.6667
d. 0.2601

36. When simplified what is 22! divided by 18!?
a. 1.2222
b. 87,780
c. 175,560
d. 2.742

37. At Governor's Cup Middle School, lockers have 7-digit passwords. If numbers in the passwords may be repeated, how many different passwords are available?
a. $10^7$
b. ${}_{10}C_6$
c. ${}_{10}P_7$
d. ${}_{10}C_7$

38. Find all solutions for X: $|3x - 7| + 5 \geq 13$.
a. $x \leq -4/3$ or $x \geq 5$
b. $x \leq -1/3$ or $x \geq 5$
c. $-4/3 \leq x \geq 5$
d. $-1/3 \leq x \geq 5$

39. Simplify the following expressions: $4x^3 + 2x^2 + 5x + 5 - (2x^3 + 6x - 4)$.
a. $2x^3 + 2x^2 - x + 9$
b. $2x^3 + 2x^2 - x - 1$
c. $2x^3 + 2x^2 + 11x + 1$
d. $6x^3 + 2x^2 + 11x - 1$

40. When you expand $(x + 5)^7$, what is the coefficient of the third term?
a. 175
b. 375
c. 525
d. 4375

41. Simplify the following algebraic fractions: $\frac{2a+16}{3} - \frac{2a-10}{3} + \frac{6a+4}{3}$.
a. $2a + 15$
b. $2a + 10$
c. $6a + 30$
d. $\frac{6a+10}{3}$

42. Rewrite the following expression using only positive exponents $\frac{12x^{-6}y^2z^{-1}}{9xy^{-5}z}$. What is the resulting denominator?
a. $4y^7$
b. $3x^7z^2$
c. $9x^7z^2$
d. $3x^7y^3z^2$

43. What is the inverse function for $g(x)=3x^3+20$?

a. $\sqrt[3]{\frac{(x-20)}{3}}$

b. $\frac{-20}{3}\sqrt[3]{x}$

c. $\sqrt[3]{\frac{20}{3}}$

d. $\sqrt[3]{.3} - 20$

44. Find the constant of proportionality if x = 16 and y = 4 follow an inverse variation.

a. 4

b. ¼

c. 64

d. 2

45. Hooke's Law says that the force exerted on a spring varies directly with the distance that the spring is stretched. If a 1.5-pound weight suspended on a spring stretches the spring 1 inch, how far will a 16-pound weight stretch it?

a. 24 inches

b. 12 inches

c. 10 2/3 inches

d. 12 2/3 inches

46. What is the slope of a line perpendicular to a line passing through the points (-4, 12) and (3, 8)?

a. -7/4

b. -4/7

c. 4/7

d. 7/4

47. Simplify the following expression: $9^{5/2}$.

a. 22.5

b. 243

c. 405

d. 2.408

48. Which of the following answers indicates the result when you combine the following radical expression:

$$\sqrt{\frac{49}{12}} - \sqrt{\frac{25}{48}}$$

a. $\sqrt{3}$

b. $3\sqrt{3}$

c. $\frac{3\sqrt{3}}{4}$

d. $\frac{4\sqrt{3}}{3}$

49. The axis of symmetry for a parabola is X = -2. The vertex for the parabola is (-2, -6). Which of the following equations could represent this parabola?

a. $y = x^{2-} + 10x + 36$
b. $y = x^{2-} - 12x + 34$
c. $y = x^2 - 4x - 2$
d. $y = x^2 + 4x - 2$

50. A quadratic function is represented by $f(x) = 2x^2 + 5x + 2$. What is the value for $f(4)$?

a. 86
b. 54
c. 52
d. 84

**Governor's Cup League Test 2 2024-2025 KEY**

1. B
2. C
3. A
4. C
5. D
6. A
7. C
8. B
9. B
10. C
11. B
12. A
13. C
14. D
15. C
16. A
17. D
18. C
19. C
20. D
21. A
22. B
23. B
24. C
25. D
26. A
27. C
28. B
29. D
30. D
31. B
32. A
33. C
34. B
35. A
36. C
37. A
38. B
39. A
40. C
41. B
42. B
43. A
44. C
45. C
46. D
47. B
48. C
49. D
50. B